

Once again, we have the power system with some generators and loads. The idea of a Load Altering Attack is that the attacker takes control of a load (or a portion of it). We call the bus of the load under attack the "victim bus" (v).

The goal of the attacker is to de-stabilise the network frequency.

Given a $f = 50$ Hz, if the frequency will go above or below a safety limit $[f - w, f + w]$, the grid will collapse.

The attacker needs to measure the frequency, and thus we have a sensor bus, where the attack places the sensor to measure the frequency. It can either be the generator or another load's bus.

According to a given attack law, the attacker will change the consumption of the victim's load. What is the mathematical law that is used? It's given by:

$$P_v^{LV} = -K_{vs}^{LG}\omega_s - K_{vs}^{LL}\phi_s$$

where P_v^{LV} is the load's variation of the victim's bus, ω_s is the frequency at generator's bus, while ϕ_s is the frequency for the sensor placed at load's bus. In the paper, though, we assume the sensor is located at the generator, thus the effective attack law is given by:

$$P_v^{LV} = -K_{vs}^{LG}\omega_s$$

K_{vs}^{LG} is the attack gain (constant), making the attack a proportional attack, and specifically the $_{vs}$ refers to v as the victim, and s the sensor.

Also v and s are to be decided by the attacker.

In the first paper, the assumptions is:

- One sensor only, while in reality the attack is more powerful by deploying more sensors across the grid.

Single-point attack, so we have only one victim's bus, and thus the attacker alters power consumption at one bus only.

Or we can have a Multi-point attack, for which the attacker controls the load at several victim's buses.

For the equations of the mathematical model, when we refer to ω , we are referring to the deviation from set points. Once again, important equations are given by:

$$M\dot{\omega} = P^M - D\omega - P^G, \quad P^M = -K^P\omega$$

When $\dot{\omega} > 0$ the machine is accelerating for example, and thus it's going too fast, and the PGC reduces the mechanical power.

If we focus on a generator, we'll have that $\dot{\omega} \sim P^M - P^G$, and the local controller controls P^M . P^G is what we're injecting in the grid, but it's not controllable by the local generator, and depends on the power flowing on the grid. It depends on load consumption and other generator's generation, and thus by the power flow equations.

When the attacker decides the consumption of the load depending on the generated power (with a sensor placed on the bus), it leads to a de-stabilising contribution.

Let's take a look now at the mathematical formulation, trying to understand in which circumstances the attack is possible and successful.

To build the model, we need to put together the power flow equations. δ is the voltage angle at generator's buses, while θ is the voltage angle at load's buses. For every generator bus, we also write swing equations. From these equations, we'll have a list of equations in (δ, ω) for generators, and a list of (θ, ϕ) for loads.

The attacker enters in the load buses.

If we take any bus i with a load, we have that P_i^L is the total power consumption at that bus. We can assume that part of the consumption is secured, and thus it cannot be attacked, and another part that is vulnerable. How the vulnerable part is changed depends on the attack law.

From the equations, we can get the matrix form. δ is the vector of all angles at generator's buses, θ is the vector of all angles at load's buses, and similar for ω and ϕ .

By analysing the matrix form in detail, for the first term of the equation the first two rows is the matrix form of $\dot{\delta}_i = \omega$, while the third two row is the matrix form of the swing equations, the fourth row instead describes the power flow equation.

The term is set equal to another matrix multiplying
$$\begin{bmatrix} \delta \\ \theta \\ \omega \\ \phi \end{bmatrix}.$$

The value of K^{LG} is the matrix of attack gains, when sensor is placed at generator's bus. In a grid, we can have several potential victim buses. Let's suppose the set of victim buses $\mathcal{V}$ is $\mathcal{V} = \{2, 5, 6\}$, so the attacker could change the load at bus 2, 5, 6. We can thus have several attack gains, depending on the victim. Let's suppose the sensor's bus is placed at bus 1. We could then have different attack scenarios. For example, if $K_{2,1}^{LG} = 0$ it means the attacker is not altering the load at bus 2.

$$\begin{cases} K_{v,s}^{LG} = 0 & \text{no attack on } v \\ K_{v,s}^{LG} > 0 & \text{attack} \end{cases}$$

So the problem is $\forall v, s$ to decide if $K_{v,s}^{LG} > 0$.

We can create a matrix where we save all the value of the gain, and we call it K^{LG} .

We also define the matrix K^{LL} , which is the matrix of attack gains when the sensor is at load bus. As we're considering that the sensor is placed at the generator bus, we have that K^{LL} is zero.

Using the last equation of the main matrix, we compute the variable ϕ as a function of all the other variables. In this model, they thus replace ϕ with the last equation.

At the end, we get this structure

$$\dot{x} = Ax + B \left(- \begin{bmatrix} 0 \\ 0 \\ K^{LG} \end{bmatrix} x \right)$$

We can then gather x , and with the attack we get $\dot{x} = Ax + BKx = (A + BK)x$. What is important here is that if we have a starting system $\dot{x} = Ax$ to say that the system is stable we need to look at the eigenvalues of the matrix A , given by λ_1 up to λ_n . We have: $Re(\lambda_i) < 0 \forall i \iff$ the system is asymptotically stable. After the attack though, the system has new equations $\dot{x} = (A + BK)x$, and the goal of the attacker (if possible) is to find an attack matrix such that at least one eigenvalue has positive real part.

Let's focus now on the vulnerability analysis. Given a grid, let's understand if this attack is possible or not. Once again, we're focusing on a single-point attack, so the attack will alter the load on one bus only.

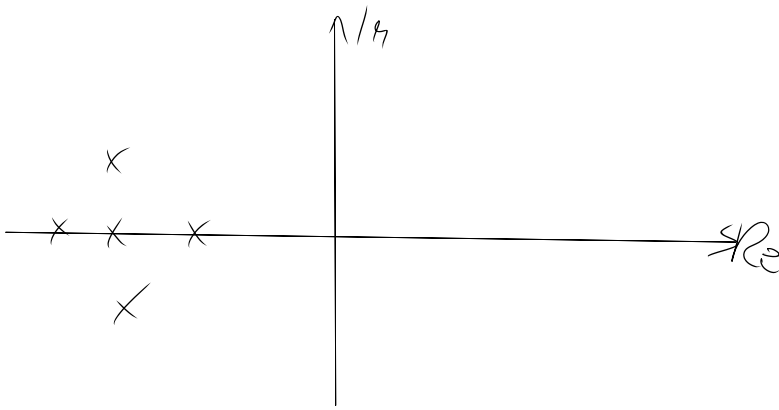
Given a grid, we can have many possible victim buses. Call $\mathcal{V}$ this set. This is a subset of all grid buses. We thus need to compute this set.

Then, define $\mathcal{S}$ the set of buses where a sensor can be placed.

Given (v, s) , can a single point attack $P_v^{LV} = -K_{vs}^{LG}\omega_s$ destabilise the grid?

We check the eigenvalues of matrix $(A - B \begin{bmatrix} 0 & 0 & K^{LG} \end{bmatrix})$.

In particular, when $K^{LG} = 0$ we have no attack, and the matrix of the system is just A , making the system stable.



All eigenvalues are on the left.

What happens if I start increasing K_{vs}^{LG} ? This will lead to a change in the eigenvalues. As K increases, each eigenvalue will move in a given way. Some of them will move towards the right, so towards instability. For a given value of K , one of these eigenvalues will cross, and will have a positive real part. We define this value of K as $(K_{v,s}^{LG})^*$. This technique is known as root locus, and with it we can compute the minimum gain that makes the system unstable.

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